

Mark Splawn

OT/IT Solutions Architect — Cloud & AI Platforms

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SUMMARY

Automation and controls engineer moving into solutions architecture, with a strong industrial / IT-OT foundation and a portfolio of self-built, production-shaped multi-cloud (GCP + AWS), event-driven serverless, and AI-agent platforms. I take ambiguous problems and turn them into systems that are buildable, operable, and affordable — designing for scalability, reliability, security, and cost. Hands-on across PLCs/SCADA, edge-to-cloud integration, and OT cybersecurity, now applied to cloud-native architecture, infrastructure as code, and CI/CD.

CORE SKILLS

Solutions Architecture & Cloud: Multi-cloud (GCP, AWS), event-driven & serverless design, Infrastructure as Code (Terraform), CI/CD with security scanning, API design, observability, secrets management & least-privilege.

AI / Agent Systems: AI agent platforms, Model Context Protocol (MCP), multi-provider model routing, retrieval-grounded generation, approval-gated autonomy.

Industrial / Controls / OT: PLC programming (Siemens S7-1500, TIA Portal), SCADA/ICS, robotics (KUKA, FANUC), industrial networking & protocols, OT cybersecurity, system integration & commissioning.

SELECTED PROJECTS

Full case studies with architecture diagrams and decision records: github.com/Markus864/Mark_Portfolio

AI Agent Platform — A reusable framework for tool-using agents: a planner over a typed tool registry with approval-gated autonomy and a provider-agnostic model mesh, plus a deployment guide.

Industrial IoT Platform — IT/OT convergence: edge telemetry bridged one-way to cloud analytics with retrieval-grounded troubleshooting, provisioned with Terraform. Validated in a sandboxed environment and presented to a company.

Serverless AI Media Platform — An event-driven pipeline (queues, durable workflows, dead-letter queues, multi-provider AI, object storage) on a serverless edge stack. Designed, built, and deployed.

Algorithmic Trading Platform — A multi-strategy engine with pluggable market-data/broker adapters, layered risk management with a kill switch, and idempotent execution. Runs live on real capital, with a backtesting + research harness.

MCP Integration Pattern — A typed MCP server that exposes an existing platform to AI agents, with a read-vs-destructive tool classification. Running in production.

PROFESSIONAL EXPERIENCE

IFA — Mechatronics Technician

Sept 2025 – Present • Automotive manufacturing

- Maintain and troubleshoot automated production systems on an automotive line, including FANUC industrial robots and Siemens PLC-controlled equipment.
- Perform diagnostics, preventive maintenance, and controls repairs across robotic cells and line equipment to maximize uptime and reliability.

Mercedes-Benz — Mechatronics Technician

Mar 2025 – Sept 2025 • Automotive manufacturing

- Maintained and troubleshoot automated production systems on an automotive line, including KUKA industrial robots and Siemens PLC-controlled equipment.
- Diagnosed and resolved electrical, pneumatic, and controls faults to minimize line downtime; performed preventive maintenance on robotic cells and conveyance.

Swisslog Logistics — Controls Technician

2019 – Mar 2025

- Led replacement of obsolete motor starters with SEW MoviMot drives, improving reliability and substantially reducing downtime.
- Directed upgrades to receiving and shipping conveyor systems, including migrating PLCs to Siemens S7-1500 controllers for seamless integration with the Warehouse Management System (WMS).
- Installed, tested, and commissioned control systems for freezer and dairy operations, maintaining performance under extreme environmental conditions.

EDUCATION

B.A.S., Advanced Manufacturing Technology — University of South Carolina Upstate • 2020 – 2022

A.S., Mechatronics — Spartanburg Community College • 2018 – 2019

CERTIFICATIONS

Google Cybersecurity • Siemens — Basics of PLCs • Siemens TIA Portal Programming I • KUKA Electrical Servicing (System Technology) • KUKA KRC • FANUC Robot Operations • FANUC PM & Lubrication • OSHA 30